



Overview

You are required to build **AssetVault**, a backend REST API system for a company's IT/Admin department that manages the complete lifecycle of internal company assets — from procurement and assignment to maintenance, transfers, and retirement.

This system will serve as the backend for the IT admin portal, enabling asset managers and department heads to track every laptop, monitor, software license, and peripheral assigned across the organization through well-documented, production-ready APIs.

Technical Guidelines (April 2025)

FRAMEWORK Spring Boot 3.3+	LANGUAGE Java 21 (LTS)	BUILD TOOL Maven — proper pom.xml
DATABASE H2 In-Memory (dev profile)	ORM Spring Data JPA + Hibernate	API DOCS SpringDoc OpenAPI 2.x — /swagger-ui.html
VALIDATION Jakarta Bean Validation	TESTING JUnit 5 + Mockito + @WebMvcTest + @DataJpaTest	LOGGING SLF4J + Logback
EXCEPTION HANDLING @RestControllerAdvice + custom exceptions	CODE QUALITY SOLID principles — DTOs mandatory	

Required Package Structure

⚠️ Entities must NOT be exposed directly in API responses — always use DTOs.

```
com.assetvault
├── config/           # SwaggerConfig, AppConfig
├── controller/      # AssetController, EmployeeController, AssignmentController
│   └── maintenance/ # MaintenanceController, LicenseController
├── service/         # Interfaces + Impl classes
├── repository/      # JPA Repository interfaces
├── model/           # JPA Entity classes
├── dto/            # Request DTOs + Response DTOs
├── exception/       # Custom exceptions + GlobalExceptionHandler
└── util/           # Constants, AssetCodeGenerator, helpers
```

Domain & Entities

Asset id, assetCode (unique, auto-generated), name, brand, model, type (LAPTOP / DESKTOP / MONITOR / KEYBOARD / MOUSE / HEADSET / PHONE / PRINTER / OTHER), status (AVAILABLE / ASSIGNED / UNDER_MAINTENANCE / RETIRED / LOST), purchaseDate, purchaseCost, warrantyExpiryDate, serialNumber, location, notes, createdAt, updatedAt	Employee id, name, email, department, designation, employeeCode, isActive, createdAt	AssetAssignment id, asset (ManyToOne), employee (ManyToOne), assignedDate, returnedDate, status (ACTIVE / RETURNED / TRANSFERRED), assignedBy, remarks, createdAt
MaintenanceRecord id, asset (ManyToOne), maintenanceType (REPAIR / SERVICE / UPGRADE / INSPECTION), description, maintenanceCost, vendor, scheduledDate, completedDate, status (SCHEDULED / IN_PROGRESS / COMPLETED / CANCELLED), createdAt	SoftwareLicense id, softwareName, licenseKey, licenseType (INDIVIDUAL / FLOATING / SITE), vendor, totalSeats, usedSeats, purchaseDate, expiryDate, assignedEmployees (ManyToOne), isActive, createdAt, updatedAt	

REST API Requirements

Asset Module — /api/v1/assets		
Method	Endpoint	Description
POST	/	Register a new asset
GET	/	Get all assets (paginated + sorted)
GET	/ {id}	Get a single asset by ID
GET	/code/{assetCode}	Get asset by its unique asset code
GET	/type/{type}	Get assets filtered by type
GET	/status/{status}	Get assets filtered by status
GET	/available	Get all currently available (unassigned) assets
GET	/department?name=	Get assets currently assigned in a department
GET	/warranty-expiring?days=	Get assets whose warranty expires within N days
GET	/warranty-expired	Get all assets with expired warranty
GET	/search?keyword=	Search assets by name, brand, or model
GET	/low-value?maxCost=	Get assets below a certain purchase cost
PUT	/ {id}	Update full asset details
PATCH	/ {id} /status?status=	Update only the asset status
PATCH	/ {id} /retire	Mark an asset as retired
PATCH	/ {id} /mark-lost	Mark an asset as lost
DELETE	/ {id}	Delete an asset record

Employee Module — /api/v1/employees		
Method	Endpoint	Description
POST	/	Register a new employee
GET	/	Get all employees (paginated)
GET	/ {id}	Get employee by ID
GET	/search?name=	Search employee by name
GET	/department?name=	Get employees by department
GET	/ {id} /assets	Get all assets assigned to an employee
GET	/ {id} /licenses	Get all software licenses assigned to an employee
GET	/ {id} /assignment-history	Get full asset assignment history of an employee
PUT	/ {id}	Update employee details
PATCH	/ {id} /deactivate	Deactivate an employee (triggers asset recall check)
DELETE	/ {id}	Delete an employee record

Assignment Module — /api/v1/assignments		
Method	Endpoint	Description
POST	/	Assign an asset to an employee
GET	/	Get all assignments (paginated)
GET	/ {id}	Get assignment by ID
GET	/active	Get all currently active assignments
GET	/asset/{assetId}	Get full assignment history for an asset
GET	/employee/{employeeId}	Get full assignment history for an employee
GET	/date-range?from=&to=	Get assignments within a date range
PATCH	/ {id} /return	Return an asset — marks assignment as RETURNED
PATCH	/ {id} /transfer?toEmployeeId=	Transfer asset from one employee to another
DELETE	/ {id}	Delete an assignment record

Maintenance Module — /api/v1/maintenance		
Method	Endpoint	Description
POST	/	Create a maintenance record for an asset
GET	/	Get all maintenance records (paginated)
GET	/ {id}	Get maintenance record by ID
GET	/asset/{assetId}	Get all maintenance records for an asset
GET	/status/{status}	Get maintenance records by status
GET	/type/{type}	Get maintenance records by type
GET	/scheduled	Get all upcoming scheduled maintenance
GET	/date-range?from=&to=	Get maintenance records within a date range
PUT	/ {id}	Update maintenance record details
PATCH	/ {id} /complete	Mark maintenance as completed
PATCH	/ {id} /cancel	Cancel a scheduled maintenance
DELETE	/ {id}	Delete a maintenance record

Software License Module — /api/v1/licenses		
Method	Endpoint	Description
POST	/	Add a new software license
GET	/	Get all licenses (paginated)
GET	/ {id}	Get license by ID
GET	/expiring-soon?days=	Get licenses expiring within N days
GET	/expired	Get all expired licenses
GET	/low-seats	Get licenses with no remaining seats
GET	/search?name=	Search licenses by software name
PUT	/ {id}	Update license details
PATCH	/ {id} /assign?employeeId=	Assign a license seat to an employee
PATCH	/ {id} /revoke?employeeId=	Revoke a license from an employee
PATCH	/ {id} /deactivate	Deactivate a license
DELETE	/ {id}	Delete a license record

Dashboard Module — /api/v1/dashboard		
Method	Endpoint	Description
GET	/summary	Total assets, assigned, available, under maintenance, retired counts
GET	/type-breakdown	Asset count grouped by type
GET	/department-assets	Asset count and value per department
GET	/warranty-alerts	Assets with warranty expiring in the next 30 days
GET	/license-alerts	Licenses expiring soon + seats exhausted
GET	/maintenance-summary	Scheduled, in-progress, and completed maintenance counts
GET	/asset-value	Total purchase cost of all active assets

⚠️ Exception Handling Requirements

All exceptions must be handled in a single `GlobalExceptionHandler` using `@RestControllerAdvice`.

AssetNotFoundException 404 Not Found Asset ID or code doesn't exist	EmployeeNotFoundException 404 Not Found Employee ID doesn't exist	AssignmentNotFoundException 404 Not Found Assignment ID doesn't exist
MaintenanceRecordNotFoundException 404 Not Found Maintenance record ID doesn't exist	LicenseNotFoundException 404 Not Found License ID doesn't exist	AssetNotAvailableException 409 Conflict Assigning an asset not in AVAILABLE status
AssetRetiredException 409 Conflict Any operation on a RETIRED asset	AssetAlreadyAssignedException 409 Conflict Asset is already actively assigned	InactiveEmployeeException 409 Conflict Assigning asset/license to a deactivated employee
NoLicenseSeatsAvailableException 409 Conflict All seats for a license are already used	LicenseExpiredException 422 Unprocessable Entity Assigning an expired software license	LicenseAlreadyAssignedException 409 Conflict Employee already holds this license
EmployeeHasActiveAssetsException 409 Conflict Deactivating employee with unreturned assets	DuplicateSerialNumberException 409 Conflict Serial number already exists in the system	MethodArgumentNotValidException 400 Bad Request Bean validation failure on request body
ConstraintViolationException 400 Bad Request Validation failure on query/path params		

```
{
  "timestamp": "2025-04-21T09:45:00",
  "status": 409,
  "error": "Conflict",
  "message": "\"Asset IPT-80123 is currently ASSIGNED and cannot be reassigned without returning it first\"",
  "path": "/api/v1/assignments"
}
```

✅ Unit Testing Requirements

Layer	Approach	What to Cover
Controller	@WebMvcTest + MockMvc + Mockito	All endpoints — success + exception scenarios
Service	@ExtendWith(MockitoExtension.class)	All business logic, edge cases, exception throws
Repository	@DataJpaTest	Custom query methods, status/type filters, date-range queries

📏 Minimum expected coverage: 80% on controller and service layers. Use `@ParameterizedTest` where multiple input scenarios apply.

📝 Logging Requirements

❌ `System.out.println` is strictly NOT allowed anywhere in the codebase.

Level	When to Log
INFO	Asset registered, assigned, returned, transferred, retired
WARN	Warranty/license expiry approaching, seat limit nearing exhaustion
ERROR	Exception caught in global handler, unexpected failures
DEBUG	Method entry/exit in service layer (dev profile only)

📖 Swagger / OpenAPI Requirements

- Swagger UI accessible at `/swagger-ui.html`
- All controllers annotated with `@Tag(name = "...", description = "...")`
- All endpoints annotated with `@Operation(summary = "...", description = "...")`
- All DTOs annotated with `@Schema(description = "...", example = "...")`
- APIs visually grouped by module in Swagger UI

🌟 Bonus (Optional but Impressive)

☐ Spring Profiles — dev (H2) and prod (MySQL/PostgreSQL)	☐ Spring Boot Actuator — /actuator/health and /actuator/info	☐ Pagination + Sorting on all list endpoints using Pageable
☐ Auto-generate unique assetCode (e.g., LPT-00123, MNT-00045)	☐ Add updatedAt auto-tracking using @LastModifiedDate	☐ Block deactivation of employee with active asset assignments

📦 Deliverable

A fully working Maven project pushed to a GitHub repository with a well-written README.md that includes setup instructions, API overview table, H2 console access steps, and the Swagger UI link.